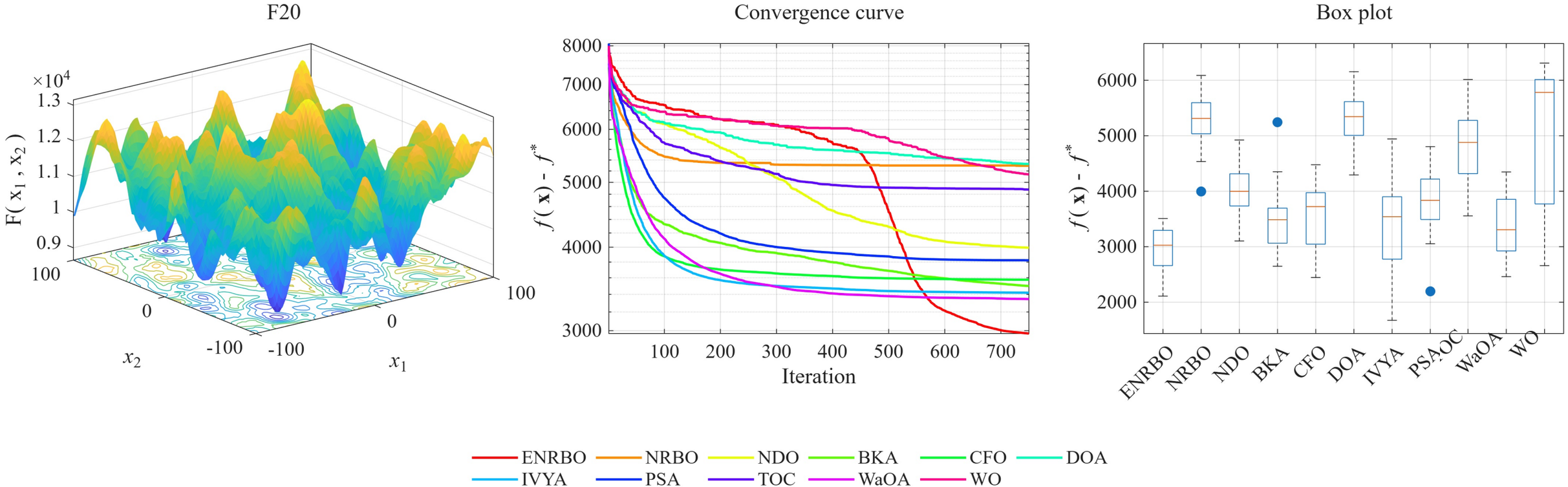

F20
×10^4
1.3
1.2
1.1
1
0.9
F( x1 , x2 )
100
0
-100
x2
-100
0
100
x1
Convergence curve
8000
7000
6000
5000
4000
3000
f( x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
6000
5000
4000
3000
2000
f( x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSAOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO